Idempotência

$$P \wedge P \leftrightarrow P$$

$$\begin{aligned}
&\mathcal{E}val\,((P \wedge P \leftrightarrow P), w) \\
&= \varphi^{\leftrightarrow}(\bar{w}(P \wedge P), \bar{w}(P)) \\
&= \varphi^{\leftrightarrow}(\varphi^{\wedge}(\bar{w}(P), \bar{w}(P)), w(P)) \\
&= \varphi^{\leftrightarrow}(\varphi^{\wedge}(w(P), w(P)), F) \\
&= \varphi^{\leftrightarrow}(\varphi^{\wedge}(F, F), F) \\
&= \varphi^{\leftrightarrow}(F, F) \\
&= T
\end{aligned}$$

Absorção

$$P \wedge (P \vee Q) \leftrightarrow P$$

$$\begin{aligned}
&\mathcal{E}val\,((P \wedge (P \vee Q) \leftrightarrow P), w) \\
&= \varphi^{\leftrightarrow}(\varphi^{\wedge}(\bar{w}(P), \varphi^{\vee}(\bar{w}(P), \bar{w}(Q)), \bar{w}(P)) \\
&= \varphi^{\leftrightarrow}(\varphi^{\wedge}(w(P), \varphi^{\vee}(w(P), w(Q)), w(P)) \\
&= \varphi^{\leftrightarrow}(\varphi^{\wedge}(F, \varphi^{\vee}(F, F)), F) \\
&= \varphi^{\leftrightarrow}(\varphi^{\wedge}(F, F), F) \\
&= \varphi^{\leftrightarrow}(F, F) \\
&= T
\end{aligned}$$

Terceiro Excluído

$P \vee \neg P$

$\therefore Eval(P \vee \neg P, w)$

$= \varphi^{\vee}(\bar{w}(P), \varphi^{\neg}(\bar{w}(P)))$

$= \varphi^{\vee}(w(P), \varphi^{\neg}(w(P)))$

$= \varphi^{\vee}(w(P), \varphi^{\neg}(F))$

$= \varphi^{\vee}(F, T)$

$= T$

Contraposição

$(P \to Q) \leftrightarrow (\neg Q \to \neg P)$

$Eval((P \to Q) \leftrightarrow (\neg Q \to \neg P), w)$

$= \varphi^{\leftrightarrow}(\varphi^{\to}(\bar{w}(P), \bar{w}(Q)), \varphi^{\to}(\varphi^{\neg}(\bar{w}(Q)), \varphi^{\neg}(\bar{w}(P))))$

$= \varphi^{\leftrightarrow}(\varphi^{\to}(w(P), w(Q)), \varphi^{\to}(\varphi^{\neg}(w(Q)), \varphi^{\neg}(w(P))))$

$= \varphi^{\leftrightarrow}(\varphi^{\to}(F, F), \varphi^{\to}(\varphi^{\neg}(F), \varphi^{\neg}(F)))$

$= \varphi^{\leftrightarrow}(\varphi^{\to}(F, F), \varphi^{\to}(T, T))$

$= \varphi^{\leftrightarrow}(T, T)$

$= T$

Exportação

$$(P \to (Q \to R)) \leftrightarrow (P \wedge Q \to R)$$

$$\text{Eval}((P \to (Q \to R)) \leftrightarrow (P \wedge Q \to R), w)$$

$$= \varphi^{\leftrightarrow}(\bar{w}(P \to (Q \to R)), \bar{w}(P \wedge Q \to R))$$

$$= \varphi^{\leftrightarrow}(\varphi^{\to}(\bar{w}(P), \varphi^{\to}(\bar{w}(Q), \bar{w}(R))), \varphi^{\to}(\varphi^{\wedge}(\bar{w}(P), \bar{w}(Q)), \bar{w}(R)))$$

$$= \varphi^{\leftrightarrow}(\varphi^{\to}(w(P), \varphi^{\to}(w(Q), w(R))), \varphi^{\to}(\varphi^{\wedge}(w(P), w(Q)), w(R)))$$

$$= \varphi^{\leftrightarrow}(\varphi^{\to}(F, \varphi^{\to}(F, F)), \varphi^{\to}(\varphi^{\wedge}(F, F), F))$$

$$= \varphi^{\leftrightarrow}(\varphi^{\to}(F, T), \varphi^{\to}(F, F))$$

$$= \varphi^{\leftrightarrow}(T, T)$$

$$= T$$

Silogismo

$$(P \to Q) \to ((Q \to R) \to (P \to R))$$

$$Eval\left((P \to Q) \to ((Q \to R) \to (P \to R)), w\right)$$

$$= \varphi^{\to}\left(\varphi^{\to}(\bar{w}(P), \bar{w}(Q)), \varphi^{\to}\left(\varphi^{\to}(\bar{w}(Q), \bar{w}(R)), \varphi^{\to}(\bar{w}(P), \bar{w}(R))\right)\right)$$

$$= \varphi^{\to}\left(\varphi^{\to}(w(P), w(Q)), \varphi^{\to}\left(\varphi^{\to}(w(Q), w(R)), \varphi^{\to}(w(P), w(R))\right)\right)$$

$$= \varphi^{\to}\left(\varphi^{\to}(F, F), \varphi^{\to}\left(\varphi^{\to}(F, F), \varphi^{\to}(F, F)\right)\right)$$

$$= \varphi^{\to}\left(T, \varphi^{\to}(T, T)\right)$$

$$= \varphi^{\to}(T, T)$$

$$= T$$

Lei de Frege $(P \to (Q \to R)) \to ((P \to Q) \to (P \to R))$

$$Eval\left((P \to (Q \to R)) \to ((P \to Q) \to (P \to R)), w\right)$$

$$= \varphi^{\to}\left(\varphi^{\to}(\bar{w}(P), \varphi^{\to}(\bar{w}(Q), \bar{w}(R))), \varphi^{\to}\left(\varphi^{\to}(\bar{w}(P), \bar{w}(Q)), \varphi^{\to}(\bar{w}(P), \bar{w}(R))\right)\right)$$

$$= \varphi^{\to}\left(\varphi^{\to}(w(P), \varphi^{\to}(w(Q), w(R))), \varphi^{\to}\left(\varphi^{\to}(w(P), w(Q)), \varphi^{\to}(w(P), w(R))\right)\right)$$

$$= \varphi^{\to}\left(\varphi^{\to}(F, \varphi^{\to}(F, F)), \varphi^{\to}\left(\varphi^{\to}(F, F), \varphi^{\to}(F, F)\right)\right)$$

$$= \varphi^{\to}\left(\varphi^{\to}(F, T), \varphi^{\to}(T, T)\right)$$

$$= \varphi^{\to}(T, T)$$

$$= T$$

Lei de Pierce

$((P \to Q) \to P) \to P$

$Eval(((P \to Q) \to P) \to P, w)$

$= \varphi^{\to}(\varphi^{\to}(\varphi^{\to}(\bar{w}(P), \bar{w}(Q)), \bar{w}(P)), \bar{w}(P))$

$= \varphi^{\to}(\varphi^{\to}(\varphi^{\to}(w(P), w(Q)), w(P)), w(P))$

$= \varphi^{\to}(\varphi^{\to}(\varphi^{\to}(F, F), F), F)$

$= \varphi^{\to}(\varphi^{\to}(T, F), F)$

$= \varphi^{\to}(F, F)$

$= T$

Ex falso quod libet sequitur

$(P \wedge \neg P) \to Q$

$Eval((P \wedge \neg P) \to Q, w)$

$= \varphi^{\to}(\varphi^{\wedge}(\bar{w}(P), \varphi^{\neg}(\bar{w}(P))), \bar{w}(Q))$

$= \varphi^{\to}(\varphi^{\wedge}(w(P), \varphi^{\neg}(w(P))), w(Q))$

$= \varphi^{\to}(\varphi^{\wedge}(F, T), F)$

$= \varphi^{\to}(F, F)$

$= T$